

# Order-Flow and Liquidity Provision<sup>☆</sup>

Álvaro Cartea<sup>a</sup>, Sebastian Jaimungal<sup>b</sup>

<sup>a</sup>*Department of Mathematics, University College London, London, UK*

<sup>b</sup>*Department of Statistical Sciences, University of Toronto, Toronto, Canada*

---

## Abstract

We show how to optimally take positions in the limit order book by placing limit orders at-the-touch when the midprice of the asset is affected by the trading activity of the market. The midprice dynamics have a short-term-alpha component which reflects how instantaneous net order-flow, the difference between the number of market buy and market sell orders, affects the asset's drift. If net-order flow is positive (negative), so short-term-alpha is positive (negative), the strategy may even withdraw from the sell (buy) side of the limit order book to take advantage of inventory appreciation (depreciation) and to protect the trading strategy from adverse selection costs.

*Keywords:* Algorithmic Trading; High Frequency Trading; Market Making; Short Term Alpha; Adverse Selection.

---

## 1. Introduction

In modern electronic markets traders provide liquidity by posting limit orders (LOs) which show an intention to buy or sell the asset and must indicate the amount of shares and price at which the investor is willing to trade. During the trading day, all orders are accumulated in the limit order book (LOB) until they find a counterparty for execution or are canceled by the investor who posted them. The counterparty is a market order (MO) which is an order to buy or sell an amount of shares, regardless of the price, which is immediately executed against limit orders resting in the LOB at the best execution prices.

Investors who specialise on providing liquidity to the market earn revenues from completing

---

<sup>☆</sup>SJ would like to thank NSERC and GRI for partially funding this work. ÁC acknowledges the research support of the Oxford-Man Institute for Quantitative Finance and the hospitality of the Finance Group at Saïd Business School.

*Email addresses:* [a.cart@ucl.ac.uk](mailto:a.cart@ucl.ac.uk) (Álvaro Cartea), [sebastian.jaimungal@utoronto.ca](mailto:sebastian.jaimungal@utoronto.ca) (Sebastian Jaimungal)

round-trip trades, buys followed by sells or sells followed by buys, which earn them the spread between their quotes. These investment strategies are exposed to prices moving in a direction that adversely affect the profitability of the strategy and expose it to risks which include inventory risk and adverse selection risk. For example, if the investor has accumulated a long position (more limit sell than limit buy orders have been filled) and prices drop, then her net worth is reduced. Similarly, if the investor sells one share of the asset to a better informed trader then it is more likely to observe and increase in the midprice of the asset, hence the expected revenue of the round-trip trade will reflect a financial loss.

Investors do not know the direction of price innovations, but order-flow, which is the number of market buy and market sell orders, generally conveys information of whether the midprice will trend up or down. As new information arrives in the market this is impounded in the midprice of the assets by traders who execute market orders (MOs) and/or traders who reposition their LOs. In general, when there is positive net order-flow (more buy than sell MOs) midprices tend to drift up, and when there is negative net order-flow (more sell than buy MOs) the midprice tends to drift down, see Cartea and Jaimungal (2014).

In this paper we show how an investor adjusts her trading strategy when she takes into account how the market's order-flow affects midprices. In addition, the investor also controls her exposure to inventory risk by restricting how much inventory she is willing to hold, long or short, and by adjusting the speed at which she expects her inventories to revert to the optimal inventory level to hold.

The rest of this paper is structured as follows. In 2 we present the model and derive the investor's optimal strategy. In Section 3 we use simulations to illustrate the performance of the strategy, and Section 4 concludes.

## 2. Investment strategy and order-flow

We model how an investor maximises terminal wealth by trading in and out of positions using LOs. The investor provides liquidity to the LOB by posting buy and sell LOs only at-the-touch, i.e. only posts LOs at the best quotes and we assume that the bid-ask spread  $\Delta$  is constant throughout the trading horizon. To formalise the problem, we list the relevant variables that we use throughout this section:

- $M^\pm = (M_t^\pm)_{0 \leq t \leq T}$  denote the counting processes corresponding to the arrival of other participants' buy (+) and sell (−) MOs which arrive at Poisson times with intensities  $\lambda^\pm$ .
- $\ell_t^\pm \in \{0, 1\}$  denote whether she is posted on the sell side (+) or buy side (−) of the LOB –  $\ell_t^\pm = 1$  when posted on that side of the LOB, and  $\ell_t^\pm = 0$  when not posted.

- $N^{\pm,\ell} = \left(N_t^{\pm,\ell}\right)_{0 \leq t \leq T}$  denote the controlled counting processes for the investor's filled sell (+) and buy (−) LOs.
- $X^\ell = (X_t^\ell)_{0 \leq t \leq T}$  denotes the investor's controlled cash process and satisfies the SDE

$$dX_t^\ell = \left(S_t + \frac{\Delta}{2}\right) dN_t^{+,\ell} - \left(S_t + \frac{\Delta}{2}\right) dN_t^{-,\ell}, \quad (1)$$

which accounts for the cash increase when a sell LO is lifted by a buy MO and the cash outflow when a buy LO is hit by an incoming sell MO.

- $Q^\ell = (Q_t^\ell)_{0 \leq t \leq T}$  denotes the investor's controlled inventory process and is given by

$$Q_t^\ell = N_t^{-,\ell} - N_t^{+,\ell}. \quad (2)$$

We further assume that if the investor is posted in the LOB, when a matching MO arrives her LO is filled with probability one. In this case,  $N_t^{\pm,\ell}$  are controlled doubly stochastic Poisson processes with intensity  $\ell_t^\pm \lambda^\pm$  – the analysis does not change substantially if we assume that conditioned on an MO arriving, the LO is filled with probability  $p \in (0, 1)$ .

The midprice of the asset  $S = (S_t)_{0 \leq t \leq T}$ , is assumed to satisfy the SDE

$$dS_t = (v + \alpha_t) dt + \sigma dW_t, \quad (3)$$

where  $W = (W_t)_{0 \leq t \leq T}$  is Brownian motion<sup>1</sup> which accounts for the reshuffle of LOs and its potential impact on the midprice, and the midprice drift is given by a long-term component  $v$  and by a short-term component  $\alpha_t$  which is a predictable zero-mean reverting process independent of  $W$ . Here the long- and short-term components are key ingredients in the profitability of the investor's strategy. For example, if the investor trades at time scales where she does not 'see' the short-term component  $\alpha_t$ , then her strategy will not only be sub-optimal, but will lose money to better informed traders – traders who are better informed will pick-off the LOs posted by the less informed liquidity provider. On the other hand, if the investor has the ability to observe  $\alpha_t$  then she will ensure that on average her strategy does not lose money to other traders and will also use this knowledge to execute more speculative trades when  $\alpha_t$  is different from zero as we shall show below.

We assume that the investor has the ability to observe order-flow and knows how this affects the drift of the midprice, see also Cartea et al. (2014) and Cartea and Jaimungal (2012). The short-term-alpha component is as a zero-mean-reverting process which jumps by

---

<sup>1</sup>The Brownian motion can safely be replaced by any martingale independent of short-term alpha component – e.g., a pure jump process so that prices remain on a discrete grid. Under this more general modelling choice, the resulting optimal strategy remains unaltered.

a random amount at the arrival times of MOs. The short-term drift jumps up when buy MOs arrive and jumps down when sell MOs arrive. As such,  $\alpha_t$  satisfies

$$d\alpha_t = -\zeta \alpha_t dt + \eta dW_t^\alpha + \epsilon_{1+M_t^+}^+ dM_t^+ - \epsilon_{1+M_t^-}^- dM_t^- \quad (4)$$

where  $\{\epsilon_1^\pm, \epsilon_2^\pm, \dots\}$  are independent random variables, with all  $+$  having identical distribution and all  $-$  having identical distribution, and with  $\mathbb{E}[\epsilon_1^\pm] < +\infty$ . They represent the size of the sell/buy MO impact on the drift of the midprice. Moreover,  $W^\alpha = (W_t^\alpha)_{0 \leq t \leq T}$  denotes a Brownian motion independent of all other processes,  $\zeta, \eta$  are positive constants, and recall that the MOs arrive at a Poisson rate of  $\lambda^\pm$ .

The investor's performance criteria is

$$H^\ell(t, x, S, \alpha, q) = \mathbb{E}_{t,x,S,\alpha,q} \left[ X_T^\ell + Q_T^\ell (S_T - (\frac{\Delta}{2} + \varphi Q_T^\ell)) - \phi \int_t^T (Q_u^\ell)^2 du \right], \quad (5)$$

and the investor imposes the restriction that inventory remains in  $Q_t^\ell \in [\underline{q}, \bar{q}]$  and the strategy's trading horizon is  $T$ .

On the right-hand side of the performance criteria are three terms. The first is the investor's terminal cash from taking positions in the market. The second is the cost that the investor incurs when employing an MO to liquidate any remaining inventory  $Q_T^\ell$  at the end of the strategy. The terminal inventory is liquidated at the midprice  $S_T$  and the costs associated to crossing the spread, liquidity taking fees, and market impact, all of which are captured by the liquidation penalty parameter  $\varphi \geq 0$ . Finally, the third term is a running inventory penalty  $\phi \int_t^T (Q_u^\ell)^2 du$  where  $\phi \geq 0$  is the target penalty parameter, see Cartea et al. (2013) for a discussion of this inventory penalty where the authors show this term stems from the investor's uncertainty in the model for the midprice.

The investor's value function is given by

$$H(t, x, S, \alpha, q) = \sup_{\ell \in \mathcal{A}} H^\ell(t, x, S, \alpha, q),$$

where the set  $\mathcal{A}$  of admissible strategies are  $\mathcal{F}$ -predictable such that the investor is not posted on the buy (sell) side if her inventory is equal to the upper (lower) inventory constraints  $\bar{q}$  ( $\underline{q}$ ).

## 2.1. The Resulting DPE

Applying the dynamic programming principle we find the investor's value function  $H$  should satisfy the dynamic programming equation (DPE)

$$\begin{aligned}
0 = & \left( \partial_t + \alpha \partial_S + \frac{1}{2} \sigma^2 \partial_{SS} - \zeta \alpha \partial_\alpha + \frac{1}{2} \eta^2 \partial_{\alpha\alpha} \right) H - \phi q^2 \\
& + \lambda^+ \max_{\ell^+ \in \{0,1\}} \left\{ \mathbb{1}_{q > \underline{q}} \mathbb{E} \left[ H(t, x + (S + \frac{\Delta}{2} \ell^+), S, \alpha + \epsilon^+, q - \ell^+) - H \right] \right. \\
& \quad \left. + \left( 1 - \ell^+ \mathbb{1}_{q > \underline{q}} \right) \mathbb{E} \left[ H(t, x, S, \alpha + \epsilon^+, q) - H \right] \right\} \\
& + \lambda^- \max_{\ell^- \in \{0,1\}} \left\{ \mathbb{1}_{q < \bar{q}} \mathbb{E} \left[ H(t, x - (S - \frac{\Delta}{2}), S, \alpha - \epsilon^-, q + \ell^-) - H \right] \right. \\
& \quad \left. + \left( 1 - \ell^- \mathbb{1}_{q < \bar{q}} \right) \mathbb{E} \left[ H(t, x, S, \alpha - \epsilon^-, q + \ell^-) - H \right] \right\} ,
\end{aligned}$$

subject to the terminal condition

$$H(T, x, S, \alpha, q) = x + q \left( S - \left( \frac{\Delta}{2} + \varphi q \right) \right) .$$

Here, the expectations are over the random jump sizes  $\epsilon^\pm$ . The various terms in the DPE carry the following interpretation. The first line in the DPE represents the drift and diffusive components of the midprice and the short-term-alpha, as well as the alpha's mean-reverting feature. The maximisation terms represent the investor's control whether to post an LO at-the-touch. The second line represents the change in value function, if the investor is posted, due to the arrival of an MO which lifts the investor's offer and simultaneously induces a jump in the short-term-alpha. The third line represents the change in the value function when an MO arrives, but the investor is not posted – in this case only the short-term-alpha jumps. Finally, the fourth and fifth lines are for the other side of the book.

The terminal condition and the form of the DPE suggests the ansatz

$$H(t, x, S, \alpha, q) = x + q S + h(t, \alpha, q) ,$$

which splits out the accumulated cash of the trading strategy up until  $t$ , the book value of the shares which are marked-to-market at the midprice, and the added value from trading in and out of positions optimally until the end of the trading horizon.

Note that  $h$  depends on time, inventory, and the short-term-alpha. Substituting this

ansatz into the above DPE we find that  $h$  satisfies

$$\begin{aligned}
0 = & \left( \partial_t - \zeta \alpha \partial_\alpha + \frac{1}{2} \eta^2 \partial_{\alpha\alpha} \right) h + \alpha q - \phi q^2 \\
& + \lambda^+ \max_{\ell^+ \in \{0,1\}} \left\{ \mathbb{1}_{q > \underline{q}} \mathbb{E} \left[ \ell^+ \frac{\Delta}{2} + (h(t, \alpha + \epsilon^+, q - \ell^+) - h(t, \alpha + \epsilon^+, q)) \right] \right\} \\
& + \lambda^- \max_{\ell^- \in \{0,1\}} \left\{ \mathbb{1}_{q < \bar{q}} \mathbb{E} \left[ \ell^- \frac{\Delta}{2} + (h(t, \alpha - \epsilon^-, q + \ell^-) - h(t, \alpha - \epsilon^-, q)) \right] \right\} \\
& + \lambda^+ \mathbb{E} [h(t, \alpha + \epsilon^+, q) - h(t, \alpha, q)] + \lambda^- \mathbb{E} [h(t, \alpha - \epsilon^-, q) - h(t, \alpha, q)] ,
\end{aligned}$$

subject to the terminal condition

$$h(T, \alpha, q) = -q \left( \frac{\Delta}{2} + \varphi q \right) .$$

The term  $\alpha q$  which appears in the first line of the equation above is responsible for making the solution to this problem be explicitly dependent on  $\alpha$ . If it were absent, then the optimal postings and  $h$  function would be independent of  $\alpha$  since the terminal conditions do not depend on  $\alpha$  and there would be no source terms in  $\alpha$ . However, it is precisely this dependence on  $\alpha$  which renders the strategy interesting and allows it to adapt to the adverse selection costs induced by the arrival of order flow.

The form of the optimising terms allow us to characterise the optimal postings in a compact form. When  $\ell = 0$  both terms that are being maximised are zero, hence, the optimal postings of the investor can be characterised succinctly as

$$\begin{aligned}
\ell^{+,*}(t, q) &= \mathbb{1} \left( \left\{ \frac{\Delta}{2} + \mathbb{E} [h(t, \alpha + \epsilon^+, q - 1) - h(t, \alpha + \epsilon^+, q)] > 0 \right\} \cap \left\{ q > \underline{q} \right\} \right) , \\
\ell^{-,*}(t, q) &= \mathbb{1} \left( \left\{ \frac{\Delta}{2} + \mathbb{E} [h(t, \alpha - \epsilon^-, q + 1) - h(t, \alpha - \epsilon^-, q)] > 0 \right\} \cap \left\{ q < \bar{q} \right\} \right) .
\end{aligned} \tag{6}$$

Here, the notation  $\mathbb{1}(\cdot)$  is the indicator function, and equals 1 if its argument holds, and zero otherwise.

### 3. Performance of the Strategy

For the purpose of focusing solely on the effect of short-term-alpha, we set the running penalty  $\phi = 0$  and the remaining model parameters are

$$\begin{aligned}
T &= 60 \text{ sec}, \quad \bar{q} = -\underline{q} = 20, \quad \lambda^\pm = 0.8333, \quad \Delta = 0.01, \quad \varphi = 0.01, \\
\eta &= 0.001, \quad \zeta = 0.5, \quad \mathbb{E}[\epsilon] = 0.005 .
\end{aligned}$$

The choice of  $\lambda^\pm$  ensures that the investor has a maximum inventory equal to 20% of the expected number of trades. With these parameters, Figure 1 shows how the optimal strategy

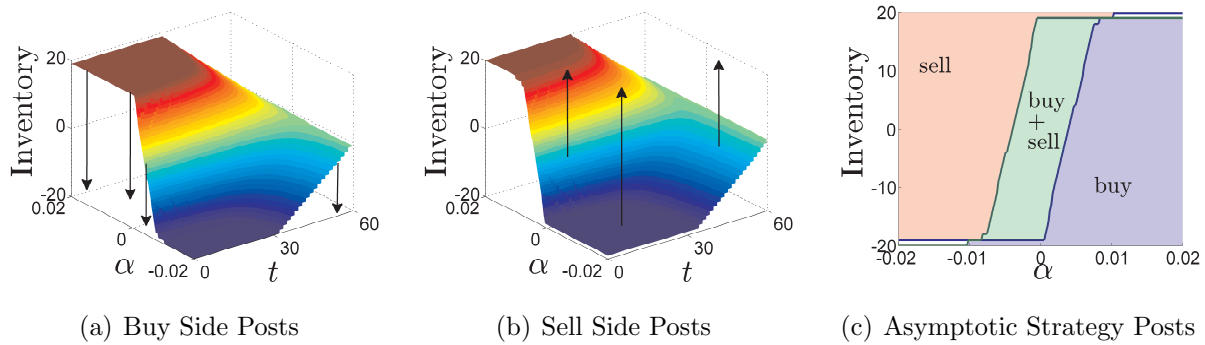


Figure 1: The optimal postings when accounting for short-term-alpha.

behaves as a function of time and short-term-alpha. The investors posts limit buys whenever her inventory is below the surface in the left panel, and she posts limit sells whenever her inventory is above the surface in the right panel. There are a number of notable features here:

1. Due to the symmetry of rates of arrival of MOs, the surfaces are mirror reflections of one another.
2. As maturity approaches her strategy becomes essentially independent of short-term-alpha, and instead induces her to sell when her inventory is positive and buy when inventory is negative. Therefore, the optimal strategy attempts to close the trading period with zero inventory.
3. The optimal strategies become time independent far from maturity.
4. Far from maturity, the investor tends to post symmetrically when short-term-alpha and inventory is close to zero. As  $\alpha$  increases, she is willing to take on inventory, but keeps posting on both sides, until  $\alpha$  becomes quite large, then she posts only buy LOs. The opposite holds when  $\alpha$  decreases.

Next, Figure 2 shows a sample path of the investors posts together with the short-term-alpha. In the left panel, the green lines highlight when (and at what price) she is posted, and the red filled circles indicate arrival of MOs that fill the investor's posts, while red unfilled circles indicate MOs that arrive when the investor was not posted. In this sample path, her postings change a total of five times and her inventory begins at zero ( $Q = 0$ ). In regime *A* she is posted only on the buy side since  $\alpha$  is large enough to suggest that purchasing the asset is worthwhile. As time evolves and she enters regime *B*,  $\alpha$  decays and she begins to post symmetrically. A buy MO arrives and lifts her offer (so that  $Q_t = -1$ ), short-term-alpha immediately jumps upwards and she removes her sell LO in regime *C*. A sell MO eventually arrives and hits her bid (so that  $Q_t = 0$ ) and immediately induces a downward jump in  $\alpha$ . Since  $\alpha$  in regime *D* is relatively small, and her inventory is zero, she posts symmetrically once again. Eventually a buy MO once again lifts her offer (so that  $Q_t = -1$ ) and induces an upward jump in  $\alpha$ . In regime *E* she now only posts on the buy side. A sequence of buy MOs arrive in this interval inducing more upward jumps in short-term-alpha, however, since

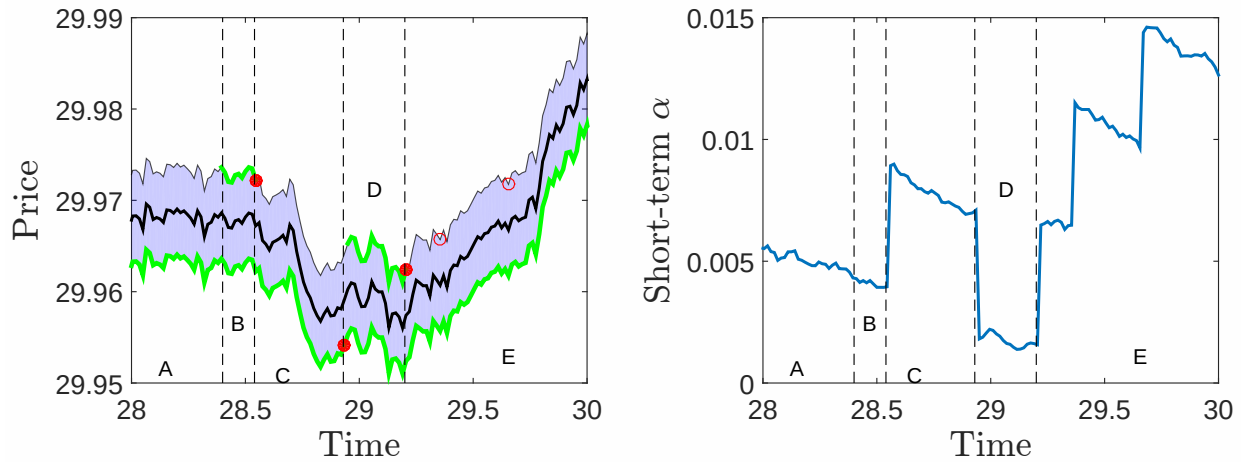


Figure 2: Sample path of the optimal strategy. Green lines show when and at what price the investor is posted. Solid red circles indicate MOs that arrive and hit/lift the posted bid/offers. Open red circles indicate MOs that arrive but do not hit/lift the investor's posts because the investor has not posted an LO. Shaded region is the bid-ask spread.

she has no sell LO posted, her inventory remains one short and she remains posted only on the buy side.

## 4. Conclusions

We show how an investor who incorporates the effect of order-flow on midprices adjusts her investment strategy to take advantage of short-term trends in the midprice and protects the strategy from adverse selection risks. Net-order-flow affects the short-term deviations of the midprice's expected growth and we show how an investor who provides liquidity to the market devises and optimal strategy to benefit from this short-term deviations which induce a short-term-alpha component in the drift of the asset.

We use simulations to show the performance of the strategy and show that, as expected, when short-term-alpha is high (low), the investor withdraws from the sell (buy) side of the LOB and is only posted on the buy (sell) side to take advantage of expected appreciation (depreciation) of the asset. Moreover, when short-term-alpha is in the neighbourhood of zero, the strategy posts two-sided quotes.

## References

Cartea, Á., R. Donnelly, and S. Jaimungal (2013). Algorithmic trading with model uncertainty. SSRN: <http://ssrn.com/abstract=2310645>.

- Cartea, Á. and S. Jaimungal (2012). Risk metrics and fine tuning of high frequency trading strategies. Forthcoming in Mathematical Finance.
- Cartea, Á. and S. Jaimungal (2013). Optimal execution with limit and market orders. Forthcoming in Quantitative Finance.
- Cartea, Á. and S. Jaimungal (2014). A closed-form execution strategy to target VWAP. SSRN eLibrary, <http://ssrn.com/abstract=2542314>.
- Cartea, Á., S. Jaimungal, and D. Kinzebulatov (2013, December). Algorithmic trading with learning. SSRN eLibrary, <http://ssrn.com/abstract=2373196>.
- Cartea, Á., S. Jaimungal, and J. Ricci (2014). Buy low, sell high: A high frequency trading perspective. SIAM Journal on Financial Mathematics 5(1), 415–444.
- Guéant, O., C.-A. Lehalle, and J. Fernandez-Tapia (2012). Dealing with the inventory risk: a solution to the market making problem. Mathematics and Financial Economics, 1–31.
- Guéant, O., C.-A. Lehalle, and J. Fernandez Tapia (2012). Optimal portfolio liquidation with limit orders. SIAM Journal on Financial Mathematics 3(1), 740–764.
- Guilbaud, F. and H. Pham (2013a). Optimal high-frequency trading in a pro rata microstructure with predictive information. Mathematical Finance, n/a–n/a.
- Guilbaud, F. and H. Pham (2013b). Optimal high-frequency trading with limit and market orders. Quantitative Finance 13(1), 79–94.
- Jaimungal, S. and D. Kinzebulatov (2014, July). Optimal Execution with a Price Limiter. Risk.